

level 1

bulldozer

a powerful machine that pushes heavy things with ease! I use wheels and levers to move my big front blade. When you push the controls, my blade lifts or lowers, showing how simple machines help do tough jobs like moving dirt and rocks!

no. of sessions: 8

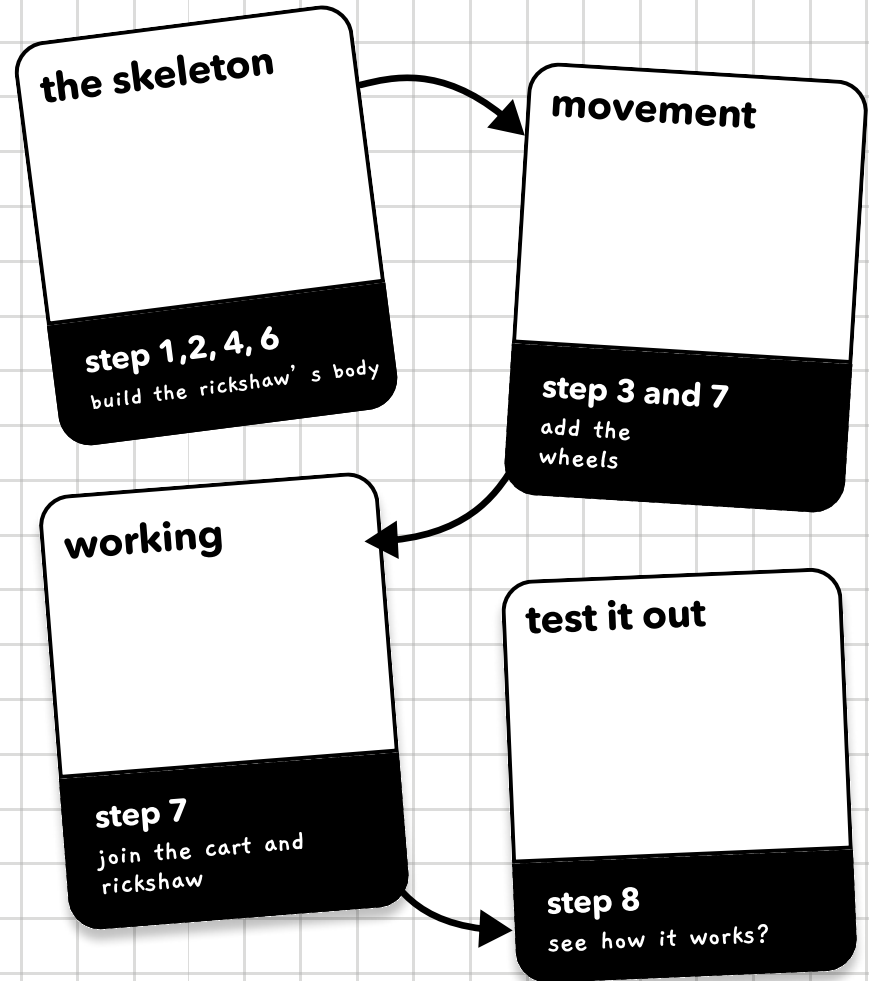
simple machine:

gears, wheels, axle & pulley

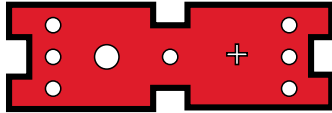
oh.



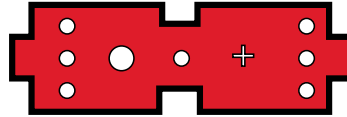
what's the goal?



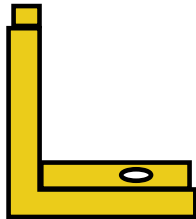
materials needed



f-f beam - x2



m-m beam - x2



L clamp - x4

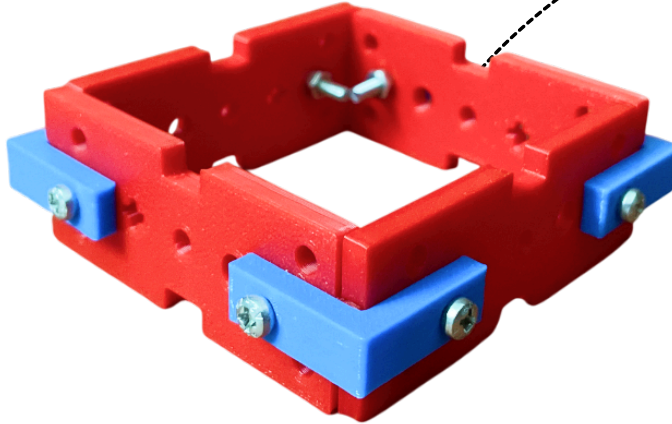


small screw - x8



nut - x8

step 1: assemble the front chassis

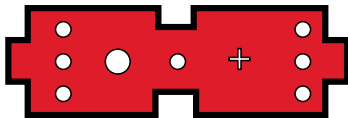


Let us start by building the chassis of the Tow Truck.

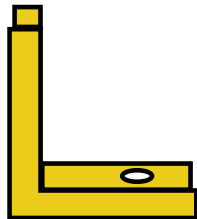
Join **2 x F-F Beams** and **2 x M-M Beams** together to form a square.

Use **4 x U Clamps** to hold them together.

materials needed



m-m beam - x1



L clamp - x1

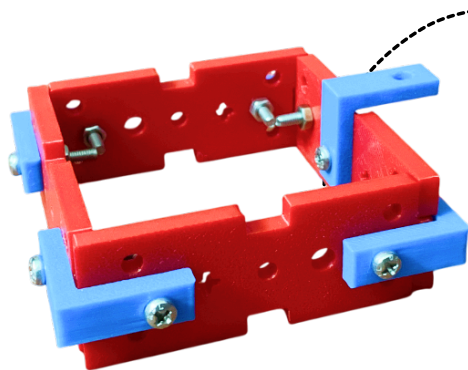


small screw - x2

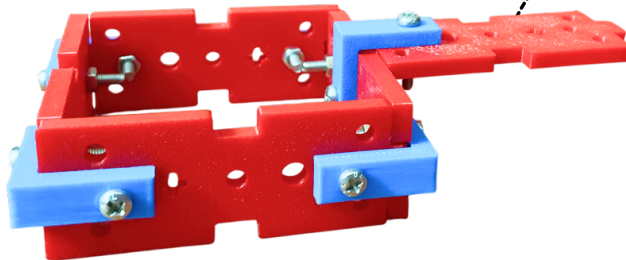


nut - x2

step 2: assemble the chassis supports

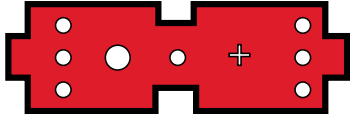


Attach a **L clamp** to the inside of the square shape.

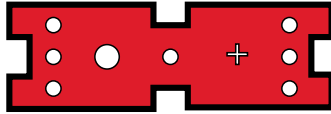


Add an **MM beam** to the other side of the L clamp.

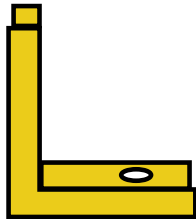
materials needed



m-m beam - x2



f-f beam - x1



L clamp - x3

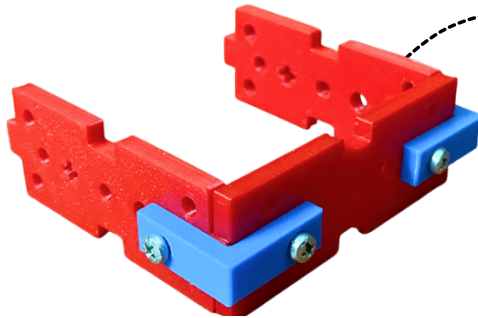


small screw - x5



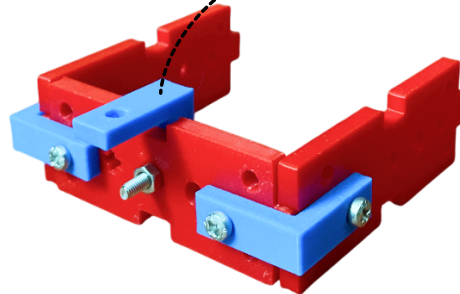
nut - x5

step 3: complete the back chassis assembly



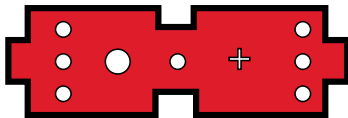
Join F-F Beam with
2 x M-M Beams together
to form a U shape.

Use **2 x L Clamps** to
hold them together.



Add a **L clamp** to
the inside of the U
shape as shown.

materials needed



m-m beam - x1

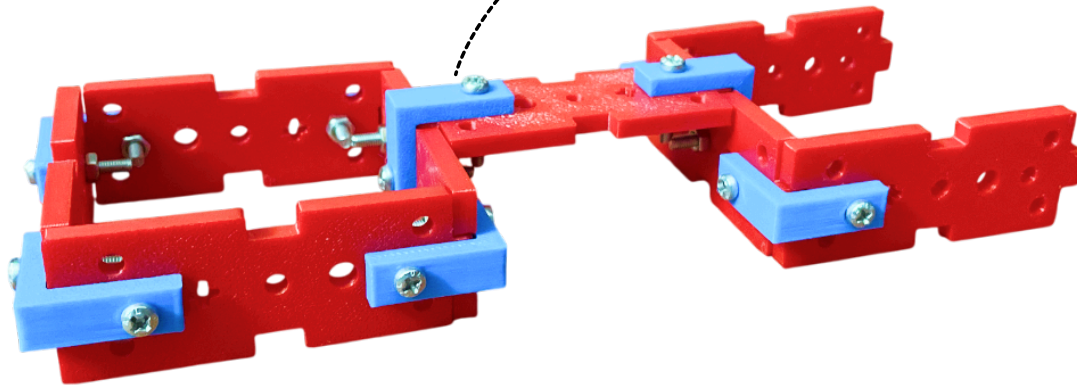


small screw - x2



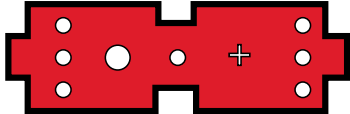
nut - x2

step 4: complete the chassis assembly

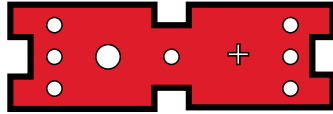


Join the square and U shape together to make the chassis of the Bulldozer.

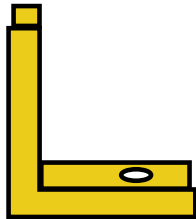
materials needed



m-m beam - x2



f-f beam - x1



L clamp - x2



I clamp - x2

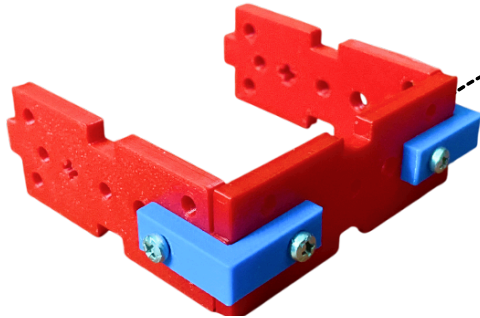


small screw - x6



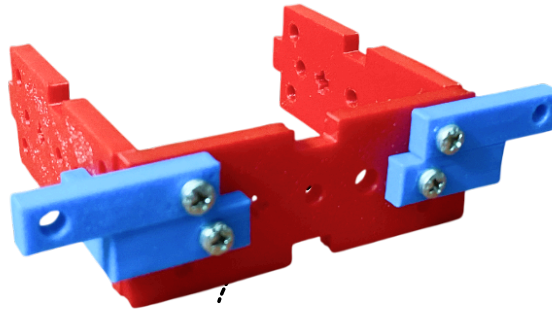
nut - x6

step 5: assemble the bulldozer claw



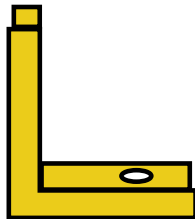
Join a F-F Beam with
2 x M-M Beams
together to form
a U shape.

Use **2 x L Clamps** to
hold them together.



Attach **2 I clamps**
to the U shape as
shown.

materials needed



L clamp - x3

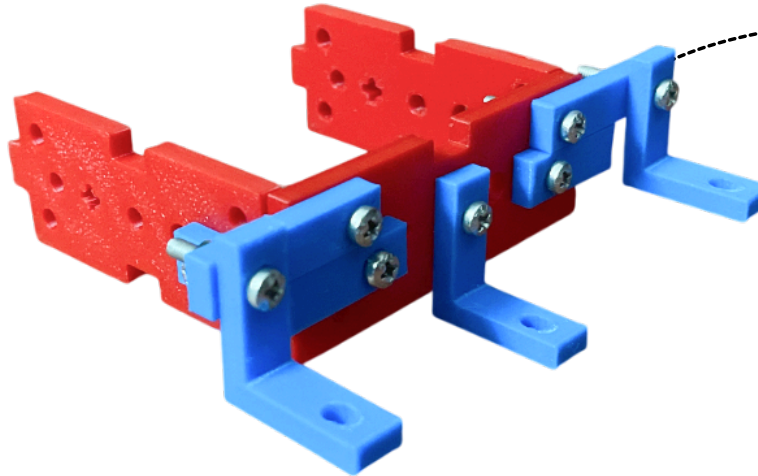


small screw - x3



nut - x3

step 6: assemble the claw supports



Take **3 L clamps** and attach it to the U shape as shown.

With this step
your bulldozer's
claw is complete.

materials needed

long axle x1

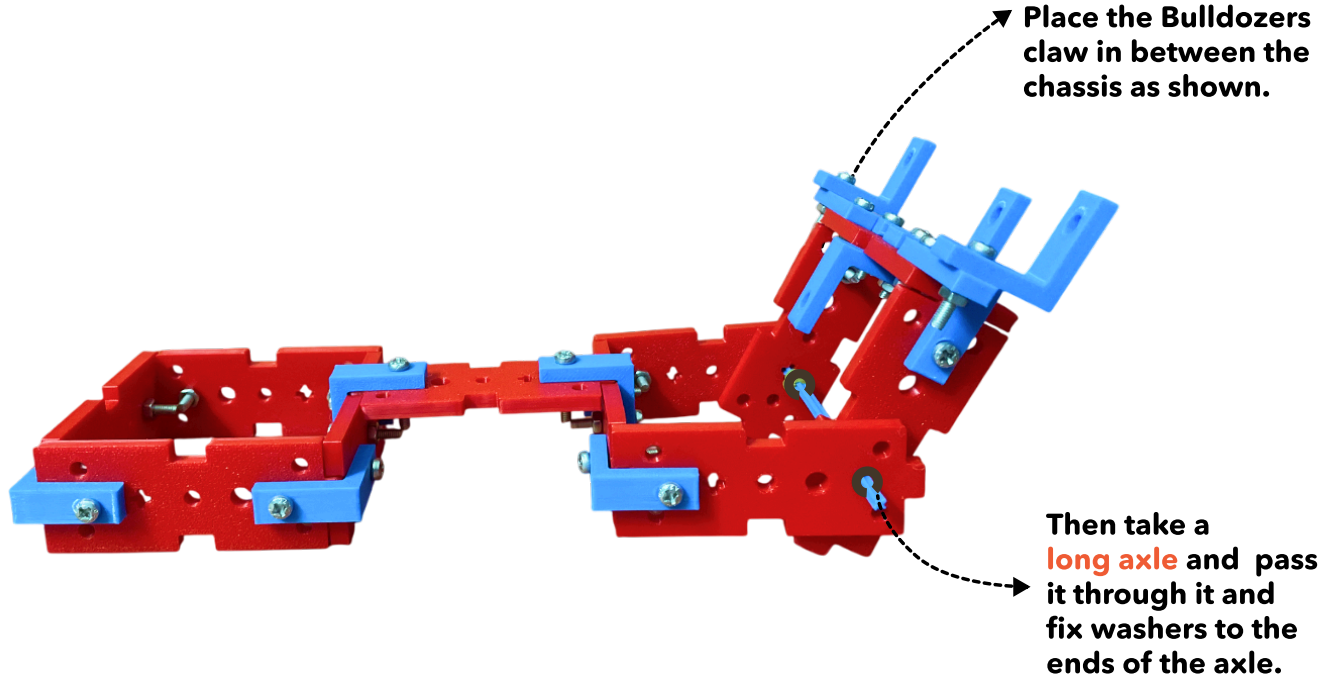


 washers x4

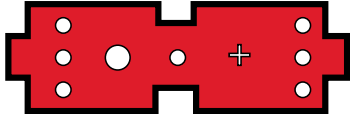


**Use washers to hold
the axle in place.**

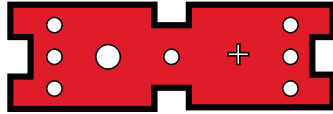
step 7: join the claw to the chassis



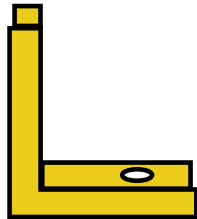
materials needed



m-m beam - x2



f-f beam - x1



L clamp - x2



I clamp - x2

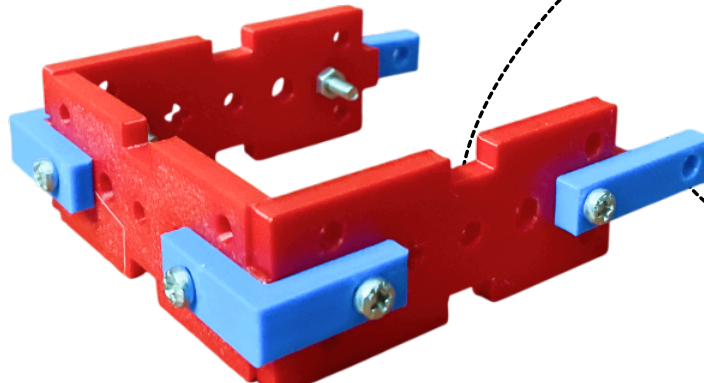


small screw - x6



nut - x6

step 8: assemble the hood

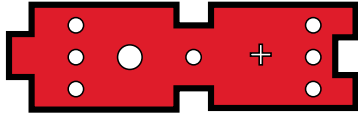


Join a **F-F Beam**
with **2 x M-M Beams**
together to form a U shape.

Use **2 x L Clamps** to
hold them together.

Attach **2 I clamps** to
the ends of the U
shape as shown.

materials needed



m-f beam - x1



small screw - x2

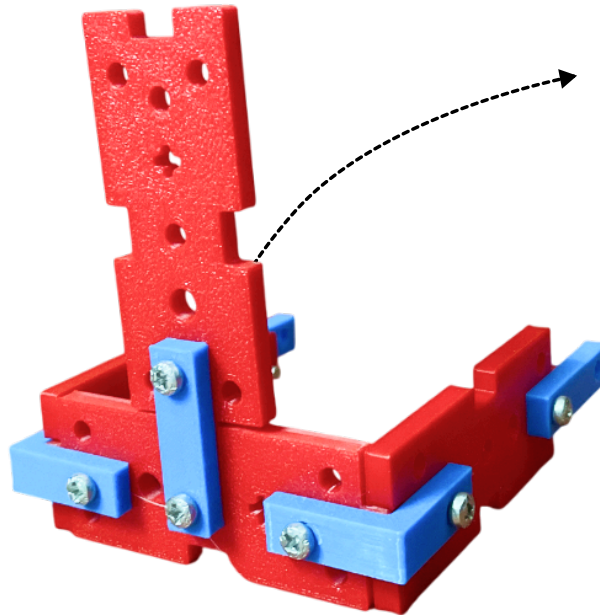


nut - x2



I clamp - x1

step 9: complete the hood assembly



Attach a **M-F beam** to the U shape as shown

With this step
the bulldozers
hood is ready.

materials needed



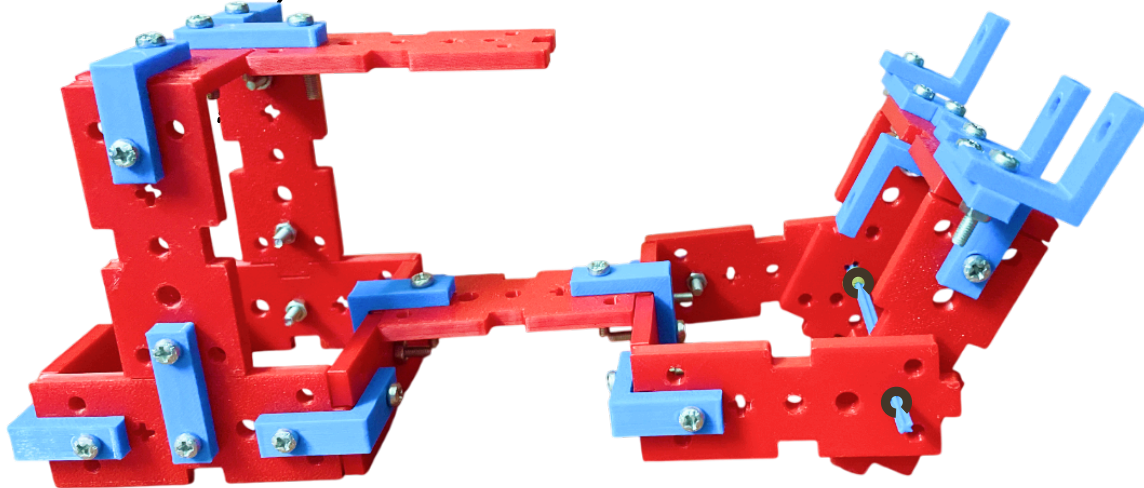
small screw - x2



nut - x2

step 10: join the hood to the chassis

Attach the hood to the back of the chassis as shown



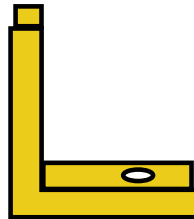
materials needed



small screw - x4

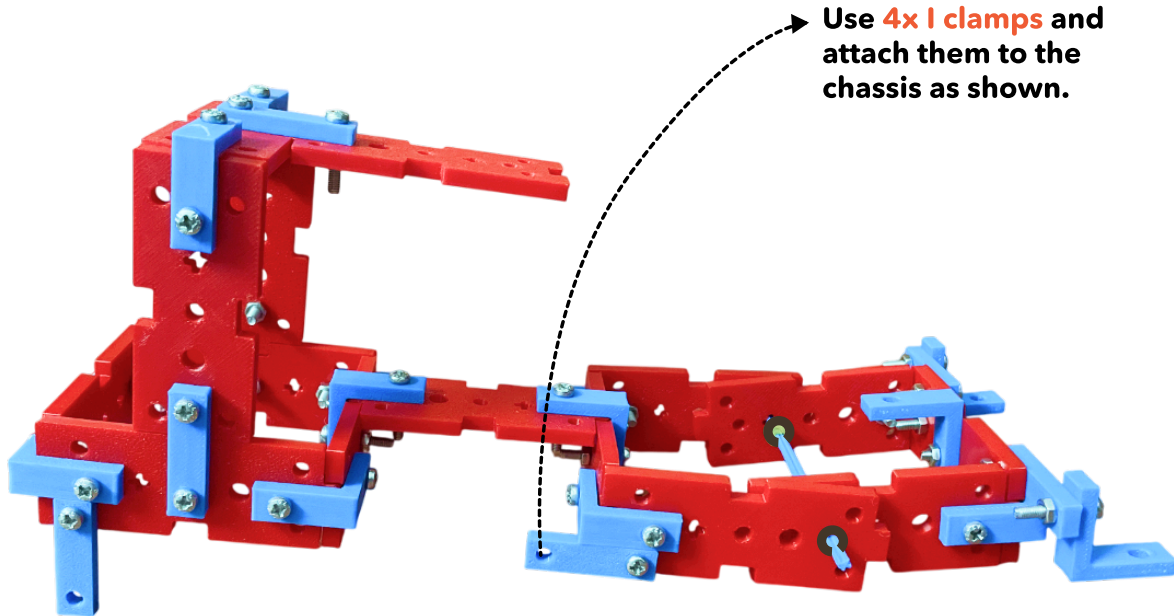


nut - x4



L clamp - x4

step 11: add the wheel supports



materials needed



wheel - x4

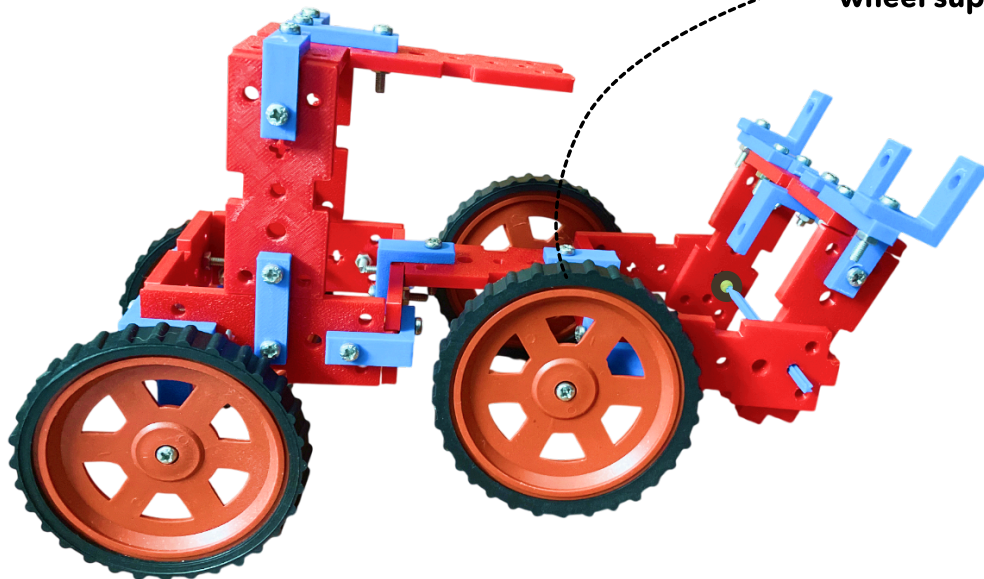


long screw - x4



nut - x4

step 12: add the wheels



Attach the 4 wheels to the wheel supports as shown.

materials needed



gear box x1



coupler - x2



I clamp - x1



long screw - x1



small screw - x1



nut - x2

step 13: assemble the gear box

Attach the **gear box** to the side of the hood using **long screws** as shown.

Add a **coupler** to the end of the gear box axle as shown.



Fix a **L clamp** to the inside of the Bulldozers claw as shown.

materials needed



coupler - x1

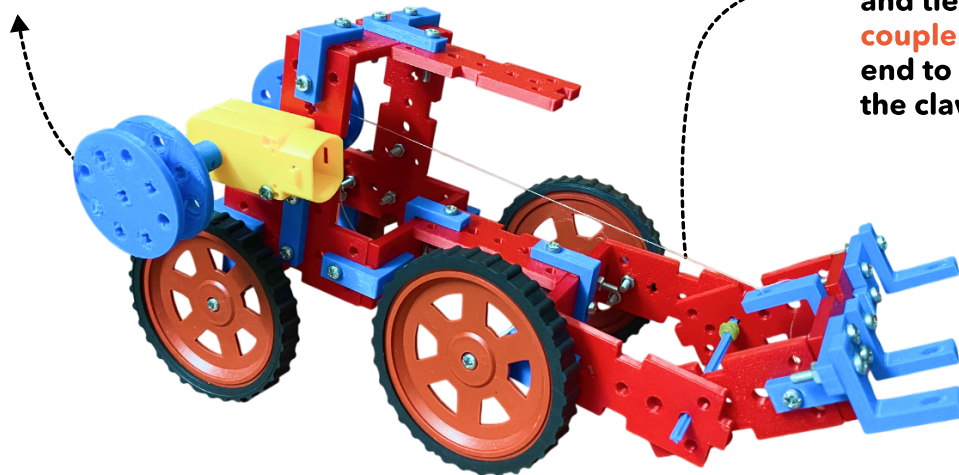


rope - 20cm - x1

step 14: complete the gearbox and pulley assembly

Add another
coupler to the other
axle of the gear box

Take a **rope of 20cm**
and tie one end to the
coupler and the other
end to the **L clamp** of
the claw.



with this step
your bulldozer
is ready for
some work!